

SUPERSTORE TASK EXPLANATION

Data Cleaning in Excel & Power BI Dashboard

1. Data Cleaning and Preprocessing – Excel

Dataset: Sample Superstore Dataset

Tool Used: Microsoft Excel

Missing Values

The dataset was checked for blank or missing values. No missing values were identified during the initial check, so no replacement or deletion was required.

Duplicate Records

Duplicate records were checked using Excel's Remove Duplicates feature. **17 duplicate records** were identified and removed, leaving **9,977 unique records** for analysis.

Data Types

- **Sales** – Numeric data type verified.
- **Quantity** – Numeric data type verified.
- **Discount** – Numeric data type verified.
- **Profit** – Numeric data type verified.

Text Consistency

The Segment, Region, Category and Sub-Category fields were checked for inconsistent text values, spelling and formatting. The values were found to be consistent and no major corrections were required.

Final Validation

After removing duplicate records and checking data types and text consistency, the dataset was validated and prepared for analysis in Power BI.

Excel Conclusion

The Sample Superstore dataset was cleaned and validated using Microsoft Excel and then used as the source for the Power BI dashboard.

2. Superstore Sales Analysis Dashboard – Power BI

Objective: To provide an interactive overview of sales performance using the Sample Superstore dataset and help users compare sales across regions, categories and sub-categories.

Key Performance Indicators

KPI	Value	Purpose
Total Sales	2.30M	Total sales generated
Total Profit	286.40K	Overall profit generated
Total Quantity	37,873	Total quantity sold

Dashboard Visuals

- **Sales by Region:** Compares sales performance across Central, East, South and West.
- **Sales by Category:** Shows sales distribution across Furniture, Office Supplies and Technology.
- **Top 10 Sub-Categories by Sales:** Highlights the sub-categories generating the highest sales.

Interactive Filter

A Region slicer is included to allow users to select a specific region or keep the selection as All. The KPI cards and visualizations update based on the selected region.

How to Read the Dashboard

Start with the KPI cards for an overall summary of sales, profit and quantity. Use Sales by Region to compare regional performance. Use Sales by Category and Top 10 Sub-Categories by Sales for more detailed analysis. The Region slicer can be used to interactively filter the dashboard.

Data Limitation

The provided dataset does not contain an Order Date or Product Name field. Therefore, monthly sales and individual product-level analysis could not be performed using the supplied dataset. Sub-Category was used for detailed product-related analysis, and no artificial data was created.

Conclusion

The Power BI dashboard provides a simple interactive view of sales performance and enables users to compare sales across regions, categories and sub-categories.